Exercise 2
Telecommunication Systems I

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Time required to complete the exercise: 5.5 hours.

Part A

In the first part of the exercise, which is mainly experimental, we will study the spectral content of baseband PAM waveforms.

A.1

Construct an SRRC pulse $\phi(t)$ with parameters $T = 10^{-3}$ sec, over $= 10$, $T_s = \frac{T}{\text{over}}$, $A = 4$, and $a = 0.5$. Using the functions `fftshift` and `fft`, compute the magnitude of the Fourier transform of $\phi(t)$, $|\Phi(F)|$, at N_f equally spaced points in the interval $\left[-\frac{F_s}{2}, \frac{F_s}{2}\right)$. Plot the energy spectral density $|\Phi(F)|^2$ versus the appropriate frequency axis using `semilogy`.

Figure:

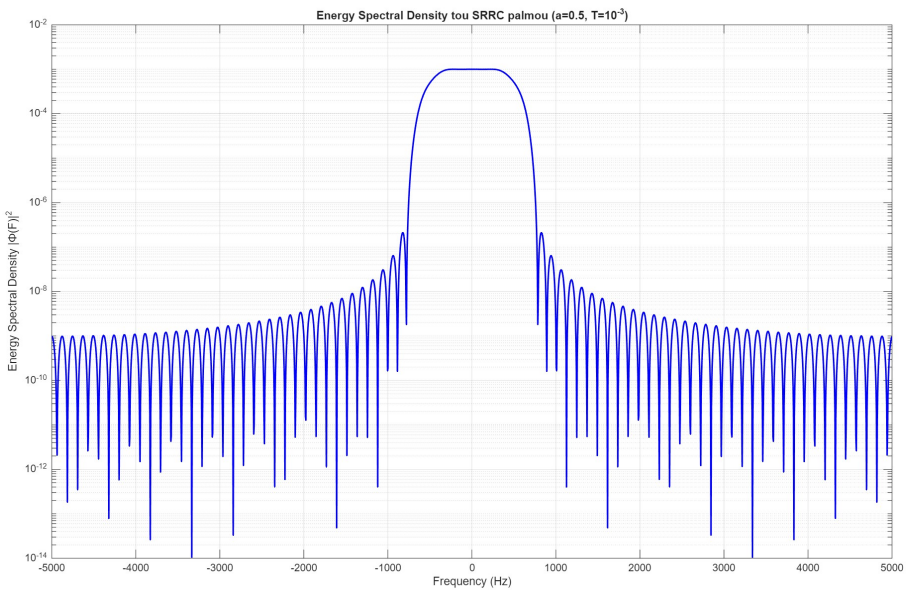

Figure 1: Energy Spectral Density of the SRRC pulse.

MATLAB Code:

```
% --- A.1 ---
T = 10^-3; over = 10; Ts = T/over; Fs = 1/Ts;
A = 4; a = 0.5; Nf = 2048;

[pulse, t] = srrc_pulse(T, over, A, a);
Freq_pulse = fftshift(fft(pulse, Nf)) * Ts;
pulse_energy = abs(Freq_pulse).^2;

f_axis = linspace(-Fs/2, Fs/2, Nf);
figure();
semilogy(f_axis, pulse_energy, 'b', 'LineWidth', 1.5);
title('Energy Spectral Density of SRRC Pulse (a=0.5, T=10^{-3})');
xlabel('Frequency (Hz)'); ylabel('Energy Spectral Density |\Phi(F)|^2');
grid on;

```

Explanation: The energy spectral density $|\Phi(F)|^2$ of the SRRC pulse was numerically calculated using FFT ($N_f = 2048$). From the plot generated with the `semilogy` function, it is observed that the majority of the pulse energy is concentrated within the theoretical bandwidth:

$$BW = \frac{1+a}{2T} = \frac{1.5}{2 \cdot 10^{-3}} = 750 \text{ Hz}$$

Beyond this frequency, the energy spectral density decays rapidly. The logarithmic y-axis scale allows for a clear visualization of the side-lobe attenuation, confirming that the SRRC filter (with $a = 0.5$) drastically limits out-of-band energy.

A.2

Generate a sequence of $N = 100$ independent and equiprobable bits $\{b_0, \dots, b_{N-1}\}$. Using the mapping $0 \to +1, 1 \to -1$, map the bits to symbols X_n. Construct the waveform:

$$X(t) = \sum_{n=0}^{N-1} X_n \phi(t - nT)$$

Assuming an infinite number of symbols, we proved that the power spectral density of $X(t)$ is $S_X(F) = \frac{\sigma_X^2}{T}|\Phi(F)|^2$.

Figure:

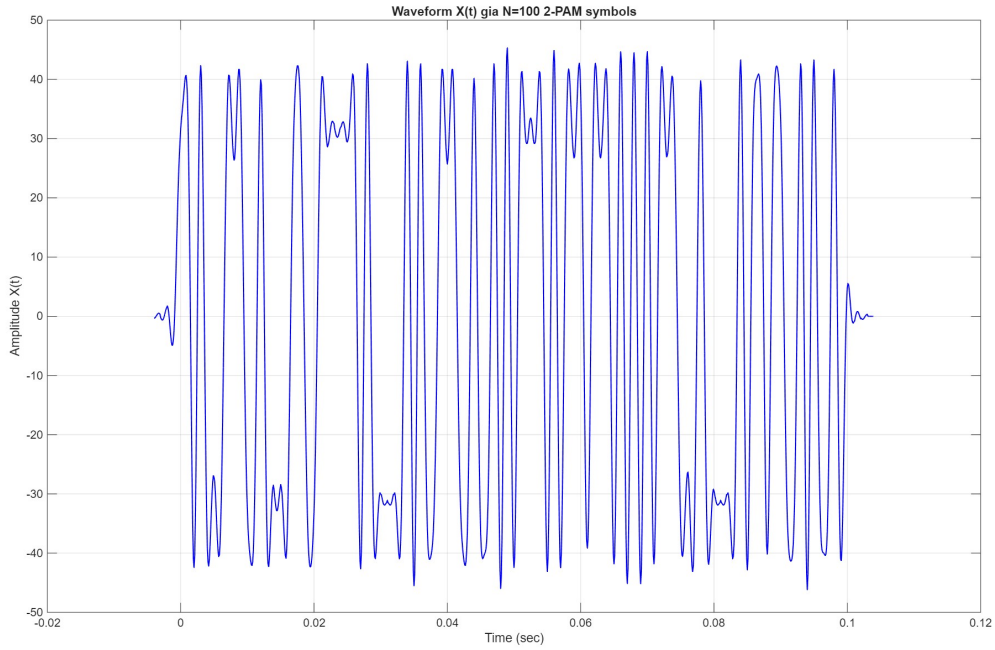

Figure 2: Constructed waveform X(t) for N=100 symbols.

MATLAB Code:

```
% --- A.2 ---
N = 100;
b = randi([0 1], N, 1);
X_n = 1 - 2*b;

X_up = zeros(length(X_n)*over, 1);
X_up(1:over:end) = X_n;
X_t = conv(X_up, pulse);
t_X = t(1) + (0:(length(X_t)-1)) * Ts;

figure(); plot(t_X, X_t, 'b');
title('Waveform X(t) for N=100 2-PAM symbols');
xlabel('Time (sec)'); ylabel('Amplitude X(t)'); grid on;
```

Explanation: A sequence of $N = 100$ random bits was generated and mapped to 2-PAM symbols ($X_n \in \{-1, +1\}$). To construct $X(t)$, the symbol sequence was upsampled by a factor of 10 and convolved with the SRRC pulse. This convolution effectively implements the summation of shifted pulses weighted by their respective symbols. Since $\sigma_X^2 = 1$ for equiprobable 2-PAM, the theoretical PSD is directly proportional to the pulse energy density scaled by $1/T$.

A.3

Using `fft` and `fftshift`, compute the periodogram of one realization of $X(t)$:

$$P_X(F) = \frac{|\mathscr{F}[X(t)]|^2}{T_{\text{total}}}$$

Estimate the power spectral density by computing numerical averages over $K = 500$ periodogram realizations.

Figures:

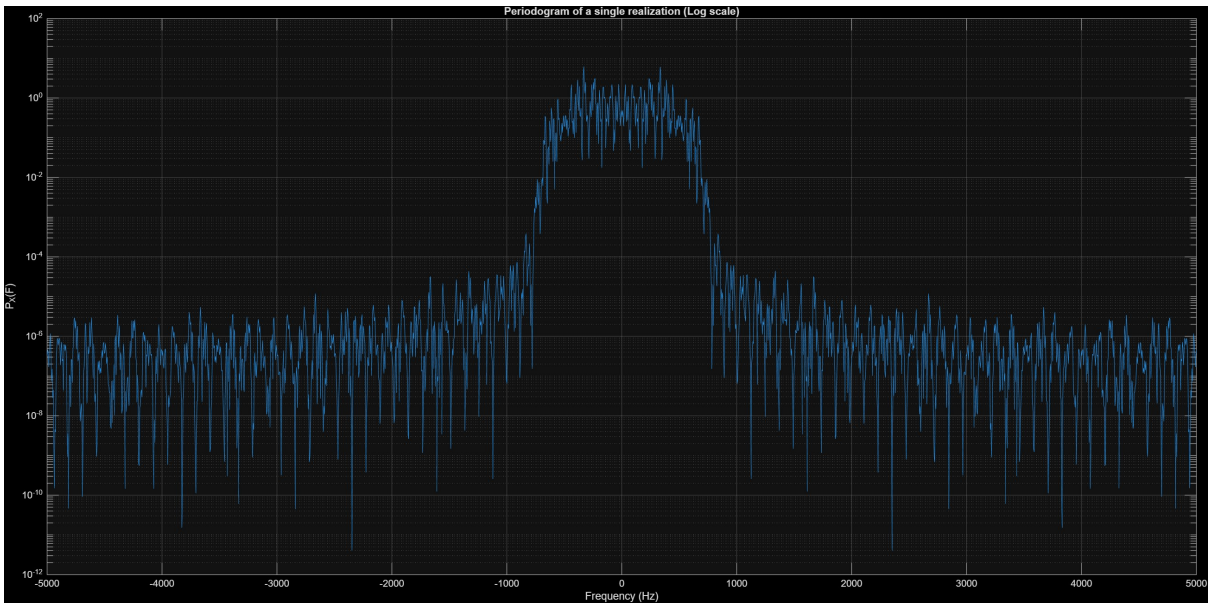

Figure 3: Periodogram of one realization (Linear and Log scale).

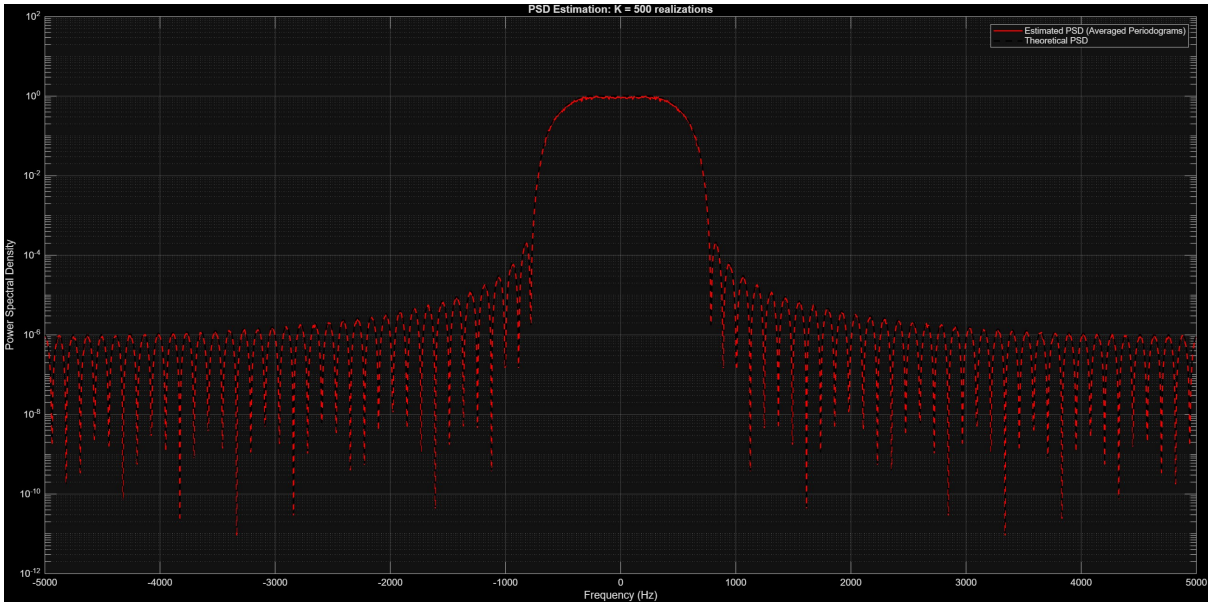

Figure 4: Estimated vs Theoretical Power Spectral Density ($K = 500$).

MATLAB Code:

```
% --- A.3 ---
T_total = length(X_t) * Ts;
Freq_X = fftshift(fft(X_t, Nf)) * Ts;
Px_F = (abs(Freq_X).^2) / T_total;

figure();
plot(f_axis, Px_F);
title('Periodogram of a single realization (Linear scale)');
xlabel('Frequency (Hz)'); ylabel('P_X(F)'); grid on;

figure();
semilogy(f_axis, Px_F);
title('Periodogram of a single realization (Log scale)');
xlabel('Frequency (Hz)'); ylabel('P_X(F)'); grid on;

K = 500;
Px_accumulated = zeros(Nf, 1);
sigma_X2 = 1;

for k = 1:K
b_k = randi([0 1], N, 1);
X_n_k = 1 - 2*b_k;

X_up_k = zeros(N*over, 1);
X_up_k(1:over:end) = X_n_k;

X_t_k = conv(X_up_k, pulse);

Freq_X_k = fftshift(fft(X_t_k, Nf)) * Ts;
Px_accumulated = Px_accumulated + (abs(Freq_X_k).^2) / T_total;
end

Px_estimated = Px_accumulated / K;

Sx_theoretical = (sigma_X2 / T) * pulse_energy;

figure();
semilogy(f_axis, Px_estimated, 'r', 'LineWidth', 1.2); hold on;
semilogy(f_axis, Sx_theoretical, 'k--', 'LineWidth', 1.5);
title(['PSD Estimation: K = ', num2str(K), ' realizations']);
xlabel('Frequency (Hz)'); ylabel('Power Spectral Density');
legend('Estimated PSD (Averaged Periodograms)', 'Theoretical PSD');
grid on;
```

Explanation: The periodogram of a single realization exhibits high variance, making it a ”noisy” estimate of the PSD. By averaging $K = 500$ independent realizations, the variance is reduced according to the Law of Large Numbers, causing the estimate to converge to the theoretical curve[cite: 32, 34]. As N increases, the frequency resolution improves and the periodogram becomes an asymptotically unbiased estimator of the true PSD.

A.4

Using the following mapping:

- $00 \longrightarrow +3$
- $01 \longrightarrow +1$
- $11 \longrightarrow -1$
- $10 \longrightarrow -3$

Construct the 4-PAM symbol sequence X_n, for $n = 0, \ldots, \frac{N}{2} - 1$. Note that if the bits are equiprobable, then the symbols X_n are also equiprobable.

Construct the waveform $X(t) = \sum_{n=0}^{\frac{N}{2}-1} X_n \phi(t - nT)$ using the same symbol period T as in question A.2. Compute the periodogram and estimate the power spectral density (PSD) by averaging periodogram realizations of $X(t)$.

Plot the experimental and the theoretical power spectral density on the same `semilogy`. How does the power spectral density of this $X(t)$ compare (in terms of bandwidth and peak magnitude) to the power spectral density of the $X(t)$ in step A.2? Explain your observations

Explanation: For equiprobable bits, the variance of the 4-PAM symbols is:

$$\sigma_X^2 = E[X_n^2] = \frac{1}{4}(3^2 + 1^2 + (-1)^2 + (-3)^2) = 5$$

Comparison with 2-PAM:

- **Bandwidth:** The bandwidth remains identical to 2-PAM because the pulse $\phi(t)$ and symbol period T are unchanged.
- **Peak Magnitude:** The PSD peak is 5 times higher ($\approx$ 7 dB) due to the increased average power ($\sigma_X^2 = 5$ vs $\sigma_X^2 = 1$).

Thus, 4-PAM transmits 2 bits per symbol in the same bandwidth but requires more power.

A.5

Repeat step A.3 by setting the symbol period to $T' = 2T$ (keep the sampling period T_s the same as in the previous steps; therefore, you must double the parameter `over`). What do you observe about the bandwidth of the waveforms in this case compared to that of the waveforms in step A.3? Can you explain the phenomenon?

Figure:

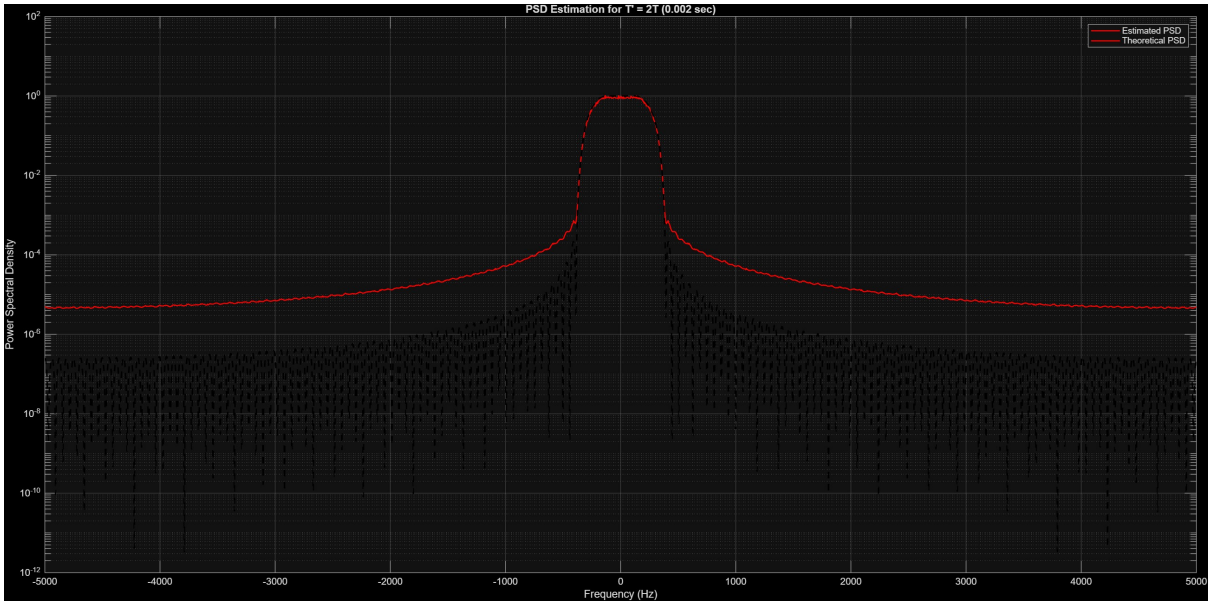

Figure 5: Estimated vs. Theoretical Power Spectral Density for $T' = 2T$.

MATLAB Code:

```
% --- A.5 ---

T_prime = 2 * T;
over_prime = 2 * over;
[pulse_A5, t_A5] = srrc_pulse(T_prime, over_prime, A, a);

Freq_pulse_A5 = fftshift(fft(pulse_A5, Nf)) * Ts;
pulse_energy_A5 = abs(Freq_pulse_A5).^2;

K = 500;
Px_accum_A5 = zeros(1, Nf);
sigma_X2 = 1;

for k = 1:K
b_k = randi([0 1], N, 1);
X_n_k = 1 - 2*b_k;

X_up_k = zeros(N * over_prime, 1);
X_up_k(1:over_prime:end) = X_n_k;

X_t_k = conv(X_up_k, pulse_A5);

T_total_A5 = length(X_t_k) * Ts;
Freq_X_k = fftshift(fft(X_t_k, Nf)) * Ts;
Px_accum_A5 = Px_accum_A5 + (abs(Freq_X_k).^2) / T_total_A5;
end
```

```
Px_est_A5 = Px_accum_A5 / K;
Sx_theo_A5 = (sigma_X2 / T_prime) * pulse_energy_A5;

figure();
semilogy(f_axis, Px_est_A5, 'r', 'LineWidth', 1.2); hold on;
semilogy(f_axis, Sx_theo_A5, 'k--', 'LineWidth', 1.5);
title(['PSD Estimation for T'' = 2T (', num2str(T_prime), ' sec)']);
xlabel('Frequency (Hz)'); ylabel('Power Spectral Density');
legend('Estimated PSD', 'Theoretical PSD');
grid on;
```

Explanation: In this section, the symbol period was increased to $T' = 2 \cdot 10^{-3}$ sec, while maintaining the same sampling frequency. The following observations were made regarding the spectral content:

- **Bandwidth Reduction:** The main lobe of the power spectral density is approximately half the width of the main lobe observed in step A.3.
- **Scaling Property:** This reduction is explained by the scaling property of the Fourier Transform, where expanding a signal in the time domain ($T' = 2T$) results in a proportional compression in the frequency domain.
- **Theoretical Bandwidth:** The new theoretical bandwidth is $BW' = \frac{1+a}{2T'} = 375$ Hz, which is exactly half of the 750 Hz bandwidth found in step A.3.
- **Peak Magnitude:** The peak magnitude of the PSD is higher compared to the previous case, as the total signal energy is concentrated within a narrower frequency range.

A.6

If you wanted to transmit data as fast as possible while using the same available bandwidth, would you choose 2-PAM or 4-PAM, and why?

If the available bandwidth is very expensive, would you choose symbol period T or $T' = 2T$, and why?

- **High Speed:** To transmit data as fast as possible in a fixed bandwidth, **4-PAM** is preferred as it doubles the bit rate ($R_b = 2/T$) compared to 2-PAM ($R_b = 1/T$).
- **Expensive Bandwidth:** If bandwidth is expensive, choosing $T' = 2T$ is preferred as it halves the spectral footprint, though it reduces the transmission speed.

Part B

In the second part of the exercise, we study the stochastic process:

$$Y(t) = X\cos(2\pi F_0 t + \Phi)$$

where $X \sim \mathcal{N}(0,1)$, $\Phi \sim \mathcal{U}[0, 2\pi)$, and X and Φ are independent random variables.

Plot, on the same plot, 5 realizations of the process. Assume $F_0 = 10$ Hz, $T_s = 10^{-2}$ sec, and `taxis` $= [0 : T_s : 5/F_0]$.

Compute the quantities $\mathcal{E}[Y(t)]$, $R_{YY}(t+\tau, t) = \mathcal{E}[Y(t+\tau)Y(t)]$, and the power spectral density $S_Y(F)$. What do you conclude?

Compute the power spectral density $S_Y(F)$

Figure:

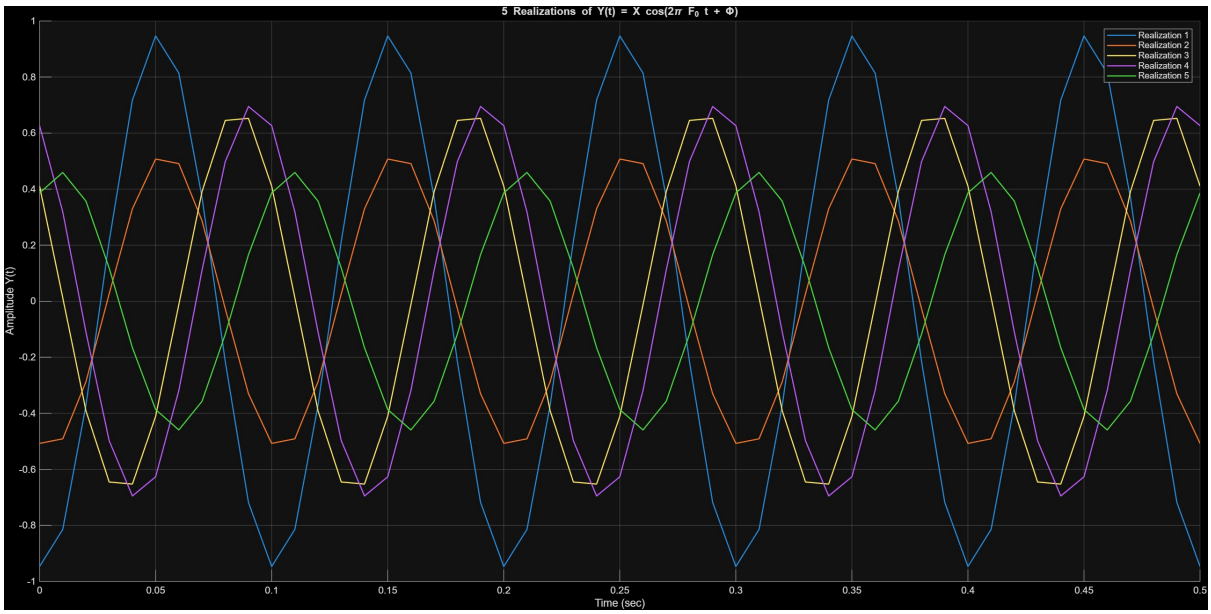

Figure 6: Five independent realizations of the stochastic process $Y(t)$.

MATLAB Code:

```
% --- B.1.1 ---

F0 = 10;
Ts = 10^-2;
t_axis = 0:Ts:5/F0;
num_realizations = 5;

figure();
hold on;
for i = 1:num_realizations
X_val = randn();
Phi_val = 2 * pi * rand();

Y_t = X_val * cos(2*pi*F0*t_axis + Phi_val);

plot(t_axis, Y_t, 'LineWidth', 1.2, 'DisplayName', ['Realization ', num2str(i)
]);
end

```

```
title('5 Realizations of Y(t) = X cos(2\pi F_0 t + \Phi)');
xlabel('Time (sec)');
ylabel('Amplitude Y(t)');
legend show;
grid on;

```

Theoretical Computations:

- **Mean Value:** Since X and Φ are independent:

$$\mathscr{E}[Y(t)] = \mathscr{E}[X]\mathscr{E}[\cos(2\pi F_0 t + \Phi)]$$

  Given $X \sim \mathscr{N}(0,1)$, we have $\mathscr{E}[X] = 0$. Therefore:

$$\mathscr{E}[Y(t)] = 0$$

- **Autocorrelation Function:** Using the identity $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$:

$$R_{YY}(t+\tau, t) = \mathscr{E}[X^2]\mathscr{E}[\cos(2\pi F_0(t+\tau) + \Phi)\cos(2\pi F_0 t + \Phi)]$$

$$R_{YY}(t+\tau, t) = \frac{1}{2}\mathscr{E}[X^2]\mathscr{E}[\cos(2\pi F_0 \tau) + \cos(2\pi F_0(2t+\tau) + 2\Phi)]$$

  Since $\mathscr{E}[X^2] = 1$ and the expected value of the second cosine term over $\Phi \sim \mathscr{U}[0, 2\pi)$ is zero, we obtain:

$$R_{YY}(\tau) = \frac{1}{2}\cos(2\pi F_0 \tau)$$

- **Power Spectral Density (PSD):** The PSD is the Fourier transform of the autocorrelation function:

$$S_Y(F) = \mathscr{F}\left\{\frac{1}{2}\cos(2\pi F_0 \tau)\right\}$$

$$S_Y(F) = \frac{1}{4}[\delta(F - F_0) + \delta(F + F_0)]$$

Explanation: Because the mean value $\mathscr{E}[Y(t)]$ is constant (zero) and the autocorrelation function $R_{YY}(t+\tau, t)$ depends only on the time difference τ, we conclude that the stochastic process $Y(t)$ is **Wide-Sense Stationary (WSS)**.